RESEARCH ARTICLE

Antiseizure Medication Polytherapies During Pregnancy and the Risk of Congenital Malformations

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Abstract

Background and Objectives

The use of antiseizure medication (ASM) polytherapy has become increasingly common. However, it is known that there is an elevated risk of major congenital malformations (MCMs) in newborns after maternal use of select ASMs. This study characterizes the risk of MCMs in infants born to women using common ASM polytherapies during the first trimester.

Methods

The study population was pregnant women enrolled in the North American AED Pregnancy Registry between 1997 and 2024. Phone interviews at enrollment, 7 months' gestation, and postpartum collected data on ASM use and maternal characteristics. ASM monotherapy or polytherapy exposure is defined by ASM or ASM combination use during the first 12 weeks of pregnancy. MCMs were confirmed by medical records and adjudicated by a blinded teratologist. The risk of MCMs was calculated among infants exposed to a specific ASM polytherapy during the first trimester of pregnancy and those exposed to lamotrigine (LTG) monotherapy. RRs and 95% CIs were estimated using logistic regression.

Results

In a population of 15,318 pregnant women taking ASM at conception, 2,348 (15.3%) used 2 or more ASMs during the first trimester, with 796 polytherapy-exposed pregnancies eligible for analysis (mean age 30 years). The most common monotherapy was lamotrigine (LTG) (n = 2,582, mean age 31 years). The most common polytherapy combination was levetiracetam-lamotrigine (LEV-LTG) (38.4%). Indication of analyzed ASM polytherapy use was largely epilepsy (98.5%). The prevalence of MCMs was 2.0% for the reference LTG monotherapy group. Compared with LTG monotherapy, the adjusted RR of MCM was 1.63 (95% CI 0.69–3.40) for levetiracetam-LTG, 0.78 (0.04–3.78) for levetiracetam-carbamazepine, 0.99 (0.05–5.29) for levetiracetam-lacosamide, 1.40 (0.22–4.97) for levetiracetam-topiramate, 1.21 (0.07–6.22) for levetiracetam-zonisamide, 3.25 (0.74–9.98) for LTG-carbamazepine, 4.54 (1.61–11.07) for LTG-topiramate, and 3.22 (0.74–9.79) for LTG-zonisamide. Among infants exposed to any ASM polytherapy, the most common MCMs were hypospadias (5 cases), ventricular septal defect (5 cases), and cleft lip/palate (4 cases). All cleft lip/palate cases occurred in combinations with either topiramate or zonisamide.

Discussion

Our findings suggest an increased risk of MCM associated with LTG-topiramate duotherapy compared with LTG monotherapy. Relative risk estimates were too imprecise for us to conclude on the teratogenicity of other analyzed polytherapy combinations.

Introduction

Multiple studies have reported comprehensive safety data on teratogenic risk of antiseizure medications (ASMs) during pregnancy, especially when used as monotherapy. Elevated risk of major congenital malformations (MCMs) in newborns was observed after maternal use of valproate, phenobarbital, and topiramate (TPM) in monotherapy compared with lamotrigine (LTG) or levetiracetam.[1-10] To avoid

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Glossary

ASM = antiseizure medication; **LCM** = lacosamide; **LEV-LTG** = levetiracetam-lamotrigine; **LMP** = last menstrual period; **LTG** = lamotrigine; **LTG-TPM** = LTG-topiramate; **MCM** = major congenital malformation; **NAAPR** = North American Anti-Epileptic Drug Pregnancy Registry; **RR** = relative risk; **RR** = Risk ratio; **TPM** = topiramate.

using teratogenic ASMs such as valproate while still achieving seizure control, alternative treatments can be considered for some patients including switching to another ASM, or lowering the dose while adding ASMs with less teratogenicity.[11] Indeed, trends of ASM use in pregnancy since 1999 have shown a growing proportion of pregnancies treated with polytherapy combinations that include LTG and levetiracetam.[12]

Studies that compared the safety of different ASM combinations in the first trimester of pregnancy established

Table 1 Characteristics of Study Participants

	Monotherapies			**Polytherapies (n = 796)**							
Categories[a]	**LTG n = 2,582**	**LEV n = 1,359**	**TPM n = 517**	**LEV-LTG n = 306 (38.4%[b])**	**LEV-CBZ n = 76 (9.5%[b])**	**LEV-LCM n = 65 (8.2%[b])**	**LEV-TPM n = 59 (7.4%[b])**	**LEV-ZNS n = 59 (7.4%[b])**	**LTG-CBZ n = 85 (10.7%[b])**	**LTG-TPM n = 83 (10.4%[b])**	**LTG-ZNS n = 63 (7.9%[b])**
Maternal age, yr, mean (SD)	31.2 (4.8)	30.9 (4.9)	29.9 (5.4)	31.1 (5.1)	30.1 (5.0)	30.7 (5.3)	29.1 (5.3)	29.0 (5.1)	29.8 (5.0)	29.1 (5.5)	29.7 (4.7)
Mother's education, n (%)											
≤ grade 12	249 (9.6)	161 (11.6)	119 (23.0)	35 (11.4)	19 (25.0)	8 (12.3)	19 (32.2)	11 (18.6)	7 (8.2)	21 (25.3)	10 (15.9)
Junior college graduate (2-y)	473 (18.3)	269 (19.3)	142 (27.5)	65 (21.2)	20 (26.3)	15 (23.1)	19 (32.2)	20 (33.9)	12 (14.1)	26 (31.3)	19 (30.2)
College graduate (4-y)	1,000 (38.7)	562 (40.4)	163 (31.5)	143 (46.7)	26 (34.2)	26 (40.0)	13 (22.0)	20 (33.9)	22 (25.9)	23 (27.7)	18 (28.6)
Postcollege	731 (28.3)	391 (28.1)	75 (14.5)	62 (20.3)	10 (13.2)	16 (24.6)	8 (13.6)	8 (13.6)	11 (12.9)	12 (14.5)	15 (23.8)
Married, n (%)	2,149 (83.2)	1,160 (83.4)	363 (70.2)	258 (84.3)	47 (61.8)	50 (76.9)	35 (59.3)	38 (64.4)	38 (44.7)	51 (61.4)	47 (74.6)
Maternal race/ethnicity, n (%)											
Non-Hispanic White	2,262 (87.6)	1,127 (81.0)	464 (90.0)	257 (84.0)	52 (68.4)	49 (75.4)	45 (76.3)	47 (79.7)	71 (83.5)	67 (80.7)	58 (92.1)
Non-Hispanic Black	52 (2.0)	52 (3.7)	9 (1.7)	11 (3.6)	8 (10.5)	3 (4.6)	5 (8.5)	1 (1.7)	6 (7.1)	4 (4.8)	0 (0.0)
Hispanic	113 (4.4)	88 (6.3)	25 (4.8)	18 (5.9)	7 (9.2)	7 (10.8)	5 (8.5)	7 (11.9)	4 (4.7)	5 (6.0)	1 (1.6)
Other	154 (6.0)	124 (8.9)	19 (3.7)	20 (6.5)	9 (11.8)	6 (9.2)	4 (6.8)	4 (6.8)	4 (4.7)	7 (8.4)	4 (6.3)
Paternal race/ethnicity, n (%)											
Non-Hispanic White	2,173 (84.2)	1,079 (77.6)	420 (81.2)	249 (81.4)	49 (64.5)	47 (72.3)	41 (69.5)	48 (81.4)	61 (71.8)	58 (69.9)	55 (87.3)
Non-Hispanic Black	88 (3.4)	82 (5.9)	25 (4.8)	24 (7.8)	10 (13.2)	3 (4.6)	7 (11.9)	3 (5.1)	7 (8.2)	5 (6.0)	1 (1.6)
Hispanic	132 (5.1)	93 (6.7)	40 (7.7)	19 (6.2)	8 (10.5)	9 (13.8)	6 (10.2)	5 (8.5)	8 (9.4)	10 (12.0)	4 (6.3)
Other	185 (7.2)	135 (9.7)	32 (6.2)	14 (4.6)	9 (11.8)	6 (9.2)	5 (8.5)	3 (5.1)	9 (10.6)	10 (12.0)	3 (4.8)
Primiparity, n (%)	703 (27.2)	389 (28.0)	136 (26.3)	85 (27.8)	25 (32.9)	11 (16.9)	17 (28.8)	13 (22.0)	21 (24.7)	19 (22.9)	15 (23.8)

Continued

Table 1 Characteristics of Study Participants *(continued)*

Categories[a]	Monotherapies			Polytherapies (n = 796)							
	LTG n = 2,582	LEV n = 1,359	TPM n = 517	LEV-LTG n = 306 (38.4%[b])	LEV-CBZ n = 76 (9.5%[b])	LEV-LCM n = 65 (8.2%[b])	LEV-TPM n = 59 (7.4%[b])	LEV-ZNS n = 59 (7.4%[b])	LTG-CBZ n = 85 (10.7%[b])	LTG-TPM n = 83 (10.4%[b])	LTG-ZNS n = 63 (7.9%[b])
Folic acid supplement use at LMP, n (%)	2065 (80.0)	1,115 (80.2)	322 (62.3)	248 (81.0)	50 (65.8)	50 (76.9)	31 (52.5)	43 (72.9)	61 (71.8)	50 (60.2)	46 (73.0)
Cigarette smoking (per d) in first trimester, n (%)											
None	2,372 (91.9)	1,301 (93.5)	442 (85.5)	279 (91.2)	64 (84.2)	61 (93.8)	54 (91.5)	53 (89.8)	75 (88.2)	69 (83.1)	56 (88.9)
>None, < (1/2) pack	98 (3.8)	53 (3.8)	27 (5.2)	13 (4.2)	6 (7.9)	4 (6.2)	2 (3.4)	4 (6.8)	3 (3.5)	5 (6.0)	3 (4.8)
≥ (1/2) pack, <1 pack	51 (2.0)	19 (1.4)	24 (4.6)	4 (1.3)	2 (2.6)	0 (0.0)	2 (3.4)	1 (1.7)	3 (3.5)	6 (7.2)	1 (1.6)
≥1 pack	49 (1.9)	13 (0.9)	20 (3.9)	10 (3.3)	4 (5.3)	0 (0.0)	1 (1.7)	0 (0.0)	3 (3.5)	2 (2.4)	1 (1.6)
Yes, but unknown	8 (0.3)	2 (0.1)	4 (0.1)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	1 (1.2)	1 (1.2)	2 (3.2)
Alcohol use (per wk) in first trimester, n (%)											
None	1992 (77.1)	1,104 (79.4)	406 (78.5)	244 (79.7)	63 (82.9)	55 (84.6)	45 (76.3)	46 (78.0)	64 (75.3)	66 (79.5)	43 (68.3)
1–5 drinks	474 (18.4)	225 (16.2)	90 (17.4)	54 (17.6)	8 (10.5)	9 (13.8)	12 (20.3)	9 (15.3)	16 (18.8)	14 (16.9)	17 (27.0)
>5 drinks	60 (2.3)	31 (2.2)	8 (1.5)	6 (2.0)	3 (3.9)	1 (1.5)	0 (0.0)	4 (6.8)	1 (1.2)	2 (2.4)	0 (0.0)
Yes, but unknown	56 (2.2)	30 (2.2)	12 (2.3)	2 (0.7)	1 (1.3)	0 (0.0)	2 (3.4)	0 (0.0)	3 (3.5)	1 (1.2)	1 (1.6)
Indication of epilepsy, n (%)	2,102 (81.4)	1,379 (99.1)	403 (77.9)	306 (100.0)	75 (98.7)	65 (100.0)	58 (98.3)	59 (100.0)	83 (97.6)	74 (89.2)	63 (100.0)
Age at first seizure, yr, mean (SD)[c]	17.6 (8.4)	18.3 (8.5)	17.0 (8.5)	14.8 (7.5)	12.4 (7.7)	14.7 (8.9)	16.2 (8.3)	13.3 (6.4)	14.8 (13.2)	14.6 (8.0)	15.0 (6.5)
Seizures during pregnancy, n (%)[c]	645 (30.7)	348 (25.6)	132 (32.8)	141 (46.1)	41 (54.7)	29 (44.6)	24 (41.3)	26 (44.1)	54 (65.1)	35 (47.3)	27 (42.9)

Abbreviations: % = percentage; CBZ = carbamazepine; LCM = lacosamide; LEV = levetiracetam; LMP = last menstrual period; LTG = lamotrigine; n = sample number; TPM = topiramate; yr = year; ZNS = zonisamide.
[a] Numbers in columns might not sum to total because of missing values.
[b] Percentage of total polytherapy pregnancy count (n = 796).
[c] Among subjects with epilepsy.

a heightened risk of MCM for polytherapies containing valproate.[13-16] Compared with valproate monotherapy, they observed lower risk of MCM for LTG-levetiracetam but not for LTG-topiramate duotherapies.[11] Although some data are available for the most common ASM combinations, because most specific polytherapy regimens are relatively infrequent treatments, more studies are necessary to evaluate their teratogenicity.[11]

To address this gap, we used data from the North American AED [antiepileptic drug] Pregnancy Registry, to estimate the risk of MCMs in offspring of individuals who were on different ASM polytherapy combinations during the first trimester of pregnancy compared with those on LTG monotherapy.

Methods

Study Design

The North American AED Pregnancy Registry, referred to as the registry moving herein forward, was established in 1996 and collects maternal and newborn health information for each enrolled woman. Women can contact the registry through 3 methods: (1) A toll-free number (1-888-233-2334), (2) Responding to an advertisement on social media (i.e., Facebook, Instagram), and (3) Submitting their name and contact information through the registry website. Women are eligible if they are pregnant at the time of initiated contact. Registry staff obtain verbal informed consent and subsequently conduct computer-assisted telephone interviews at 3 times: enrollment, 7 months' gestation, and 8–12 weeks

Table 2 Risk, Crude Relative Risk, and Adjusted Relative Risk of Major Congenital Malformations (MCM) Identified Among Infants Who Had Been Exposed to a Specific ASM Polytherapy Combination During the First Trimester Compared With the Active Comparator Group of Lamotrigine Monotherapy-Exposed Infants: North American AED Pregnancy Registry 1997–2024

	Cohort-wide (n = 4,767)				Participants with epilepsy only (n = 4,272)			
ASM regimen	Infant count, n	Major congenital malformations, n (%[a]) (95% CI)	Unadjusted RR (95% CI)	Adjusted RR[b] (95% CI)	Infant count, n	Major congenital malformations, n (%[a]) (95% CI)	Unadjusted RR (95% CI)	Adjusted RR[b] (95% CI)
LTG monotherapy	2,582	52 (2.0) (1.5–2.6)	Reference	Reference	2,102	41 (2.0) (1.4–2.6)	Reference	Reference
LEV monotherapy	1,389	26 (1.9) (1.3–2.7)	0.93 (0.57–1.48)	0.99 (0.58–1.66)	1,387	26 (1.9) (1.3–2.7)	0.97 (0.58–1.58)	0.99 (0.57–1.71)
LEV-LTG	306	8 (2.6) (1.3–5.1)	1.31 (0.57–2.62)	1.63 (0.69–3.40)	306	8 (2.6) (1.3–5.1)	1.35 (0.58–2.76)	1.54 (0.64–3.29)
LEV-CBZ	76	1 (1.3) (0.2–7.1)	0.65 (0.04–3.02)	0.78 (0.04–3.78)	75	1 (1.3) (0.2–7.2)	0.68 (0.04–3.19)	0.76 (0.04–3.75)
LEV-LCM	65	1 (1.5) (0.3–8.2)	0.76 (0.04–3.55)	0.99 (0.05–5.29)	65	1 (1.5) (0.3–8.2)	0.79 (0.04–3.70)	0.79 (0.04–4.35)
LEV-TPM	59	2 (3.4) (0.9–11.5)	1.71 (0.28–5.68)	1.40 (0.22–4.97)	58	2 (3.4) (1.0–11.7)	1.80 (0.29–6.04)	1.72 (0.27–6.31)
LEV-ZNS	59	1 (1.7) (0.3–9.0)	0.84 (0.05–3.93)	1.21 (0.07–6.22)	59	1 (1.7) (0.3–9.0)	0.87 (0.05–4.09)	1.07 (0.06–5.67)
LTG-CBZ	85	4 (4.7) (1.8–11.5)	2.40 (0.71–6.05)	3.25 (0.74–9.98)	83	4 (4.8) (1.9–11.7)	2.55 (0.75–6.50)	3.38 (0.76–10.6)
LTG-TPM	83	6 (7.2) (3.4–14.9)	3.79 (1.42–8.45)[c]	4.54 (1.61–11.07)[c]	74	5 (6.8) (2.9–14.9)	3.64 (1.23–8.71)[c]	4.49 (1.41–12.0)[c]
LTG-ZNS	63	3 (4.8) (1.6–13.1)	2.43 (0.58–6.86)	3.22 (0.74–9.79)	63	3 (4.8) (1.6–13.1)	2.51 (0.60–7.17)	3.05 (0.69–9.59)

Abbreviations: % = percentage; CBZ = carbamazepine; LCM = lacosamide; LEV = levetiracetam; LTG = lamotrigine; n = sample number; RR = relative risk; TPM = topiramate; ZNS = zonisamide.
[a] Percentage of infant with MCM count in each monotherapy or polytherapy group.
[b] Adjusting for maternal age, marital status, and calendar year of last menstrual period.
[c] Relative risk with 95% CI > 1.

after the expected date of delivery. With the participant's written permission, medical records of the infant and the mother are obtained from the infant's pediatrician, family practitioner, neurologist, psychiatrist, or other treating physicians. Collected data include demographics, ASM information (dosage, frequency, and medical indication information for each ASM taken), signs and symptoms of epilepsy, maternal obstetrical history (e.g., gestational diabetes and/or hypertension), maternal habits (e.g., alcohol use, cigarette smoking status, recreational drug use, cannabidiol use), potential maternal teratogenic exposures (e.g., use of acne medication isotretinoin [Accutane]), folate and multivitamin supplementation, fertility history, and family history of birth defects and/or epilepsy. The postpartum interview in addition asks for obstetrical and neonatal outcomes (e.g., MCMs, results of genetic testing, type of delivery).

Study Population

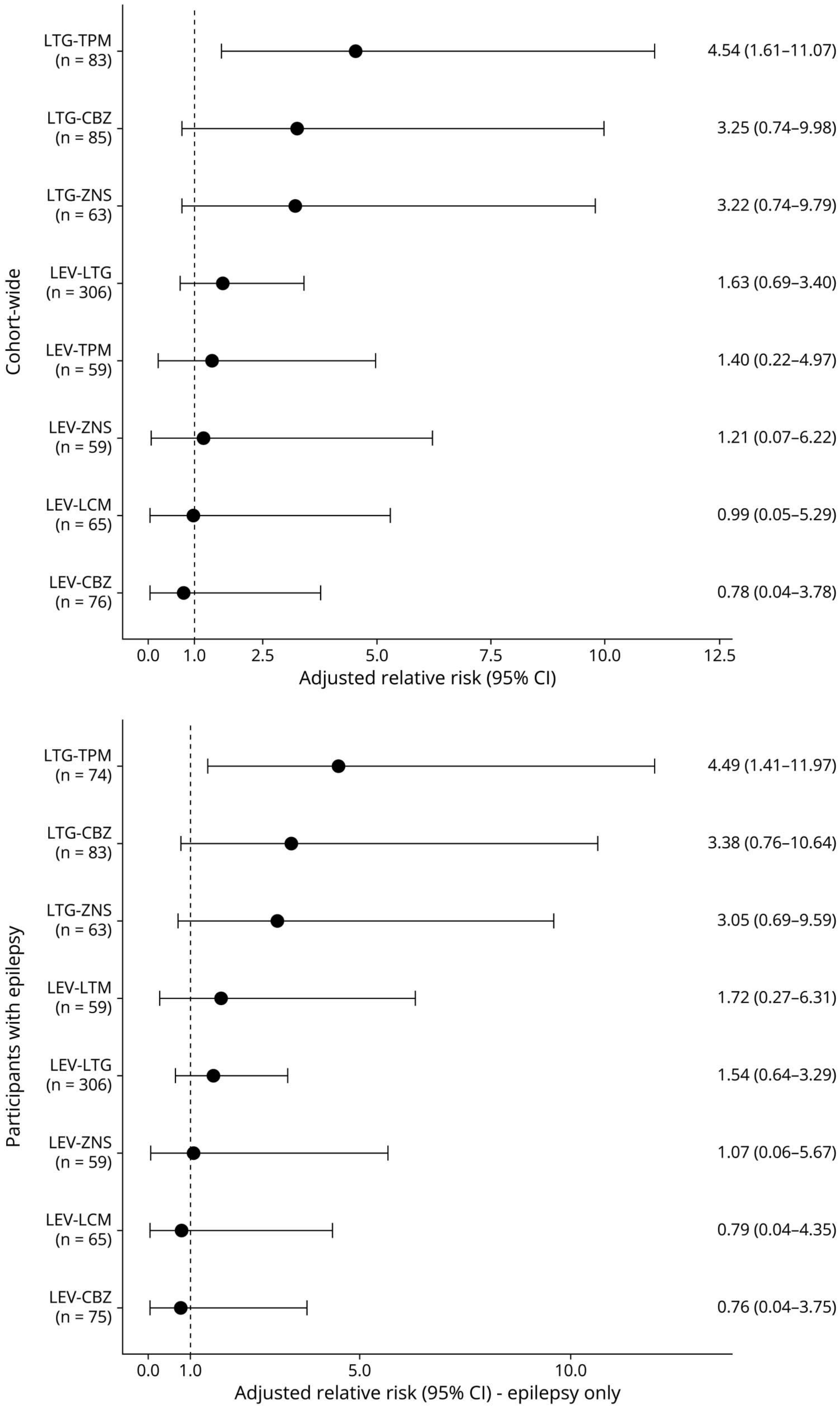
Enrolled women were included in the analysis if they completed all 3 interviews, were taking some ASM during the first 12 weeks of pregnancy, and had a liveborn, stillborn (>20 weeks' gestational age) infant, or a pregnancy termination because of a fetal abnormality. Units of analysis are pregnancies. "Pure" prospective participants are women who enrolled in the registry before any prenatal screening such as nuchal translucency screening, chorionic villus sampling, amniocentesis, or an ultrasound after 15 weeks' gestation. "Traditional" prospective participants may enroll with existing knowledge of fetal status through screening tests. Only polytherapy-exposed groups with 50 or more pregnancies were analyzed.

Exposure Definition

Women were considered exposed to a polytherapy combination if they used 2 or more ASMs simultaneously or in sequence at any time during the first 12 weeks of pregnancy (defined as time after last menstrual period (LMP)). "Monotherapy" is defined as exposure to only 1 ASM during the same pregnancy period.

An internal active-comparator group of LTG monotherapy users in the first 12 weeks of pregnancy was used as reference because (1) LTG monotherapy is the most common ASM monotherapy regimen in the registry, (2) LTG monotherapy

Figure 1 Adjusted Relative Risk of Major Congenital Malformations Identified Among Infants Who Had Been Exposed to a Specific ASM Polytherapy Combination During the First Trimester Compared With the Active Comparator Group of Lamotrigine Monotherapy-Exposed Infants, in a Cohort-wide and an Epilepsy Participant-Only Analysis



% = percentage; CBZ = carbamazepine; LCM = lacosamide; LEV = levetiracetam; LTG = lamotrigine; RR = relative risk; TPM = topiramate; ZNS = zonisamide.

Table 3 Risk, Crude Relative Risk, and Adjusted Relative Risk of Major Congenital Malformations (MCM) Identified Among Infants Who Had Been Exposed to Lamotrigine- or Levetiracetam-Polytherapy Combination During the First Trimester Compared With the Active Comparator Group of Lamotrigine Monotherapy-Exposed Infants: North American AED Pregnancy Registry 1997–2024

	Cohort-wide (n = 3,378)				Participants with epilepsy only (n = 2,885)			
ASM regimen	Infant count, n	Major congenital malformations, n (%[a]) (95% CI)	Unadjusted RR (95% CI)	Adjusted RR[b] (95% CI)	Infant count, n	Major congenital malformations, n (%[a]) (95% CI)	Unadjusted RR (95% CI)	Adjusted RR[b] (95% CI)
LTG monotherapy	2,582	52 (2.0) (1.5–2.6)	Reference	Reference	2,102	41 (2.0) (1.4–2.6)	Reference	Reference
LEV-LTG	306	8 (2.6) (1.3–5.1)	1.31 (0.57–2.62)	1.63 (0.69–3.40)	306	8 (2.6) (1.3–5.1)	1.35 (0.58–2.76)	1.54 (0.64–3.29)
LEV-non-LTG[c]	259	5 (1.9) (0.8–4.4)	0.96 (0.33–2.20)	1.11 (0.37–2.66)	257	5 (1.9) (0.7–4.7)	1.00 (0.34–2.32)	1.08 (0.36–2.69)
LTG-non-LEV[c]	231	13 (5.6) (3.3–9.4)	2.90 (1.49–5.25)[d]	3.61 (1.74–7.01)[d]	220	12 (5.5) (3.0–9.6)	2.90 (1.44–5.44)[d]	3.46 (1.59–7.01)[d]

Abbreviations: % = percentage; LEV = levetiracetam; LTG = lamotrigine; n = sample number; RR = relative risk.
[a] Percentage of infant with MCM count in each monotherapy or polytherapy group.
[b] Adjusting for maternal age, marital status, and calendar year of last menstrual period.
[c] Non-LTG and non-LEV ASMs include carbamazepine, lacosamide, topiramate, and zonisamide.
[d] Relative risk with 95% CI > 1.

use during pregnancy has not been shown to increase risk of MCMs in newborn compared with non–ASM-exposed pregnancies,[3] and (3) use of an active-comparator reference minimizes confounding by indication.

Outcome Definitions

Outcomes of interest were MCMs, defined as structural abnormalities with surgical, medical, or cosmetic importance.[17] The following physical features were excluded in the registry's outcome definition: minor anomalies, birth marks, deformations, physiologic features of prematurity (e.g., patent ductus arteriosus), any MCM detected using prenatal ultrasound but not by an examining physician at birth or postnatal imaging, subclinical malformations detected on imaging (e.g., muscular ventricular septal defects), and MCMs because of a chromosomal or monogenic abnormality. If the enrolled participant consented to medical record release, the received neonatal medical records are reviewed by the teratologist (F.A.H.) while blinded to exposure status.

Analysis

Demographic, clinical, and perinatal characteristics of women exposed to specific ASM polytherapy combinations were described. The risk of MCMs for each ASM polytherapy combination group was compared with the risk in the LTG monotherapy group. A post hoc analysis of the relative risk (RR) of MCMs of LEV monotherapy compared with LTG monotherapy was implemented because of LEV being present in a large proportion of analyzed polytherapy combinations. In addition, we analyzed RR of MCMs with a reference group of TPM monotherapy users in the first 12 weeks of pregnancy to address the teratogenic signals between TPM monotherapy and TPM polytherapy combinations. Crude RR and their 95% CIs were estimated. Multivariable logistic regression was used, given the rare outcome of interest, to estimate the adjusted RRs of MCMs of polytherapy combination groups compared with LTG-monotherapy or TPM-monotherapy reference group. Potential confounders were added one at a time to the regression model and only those that affected the estimates were retained. The following confounders were adjusted for maternal age, marital status, and calendar year of LMP.[1] Complete case analysis was conducted with participants who completed the postpartum interview.

Several sensitivity analyses were conducted. First, to mitigate potential differential enrollment based on results of prenatal screening tests, analysis was restricted to pure prospective participants. Second, analysis was restricted to women with epilepsy to further adjust for ASM indication.

RStudio version 4.4.1 was used for all statistical analyses. This manuscript was written in alignment with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE; version 4) guidelines (eTable 1).

Standard Protocol Approvals, Registrations, and Patient Consents

Informed consent was obtained verbally at enrollment. This study has been approved annually by the Institutional Review Board of the Mass General Brigham.

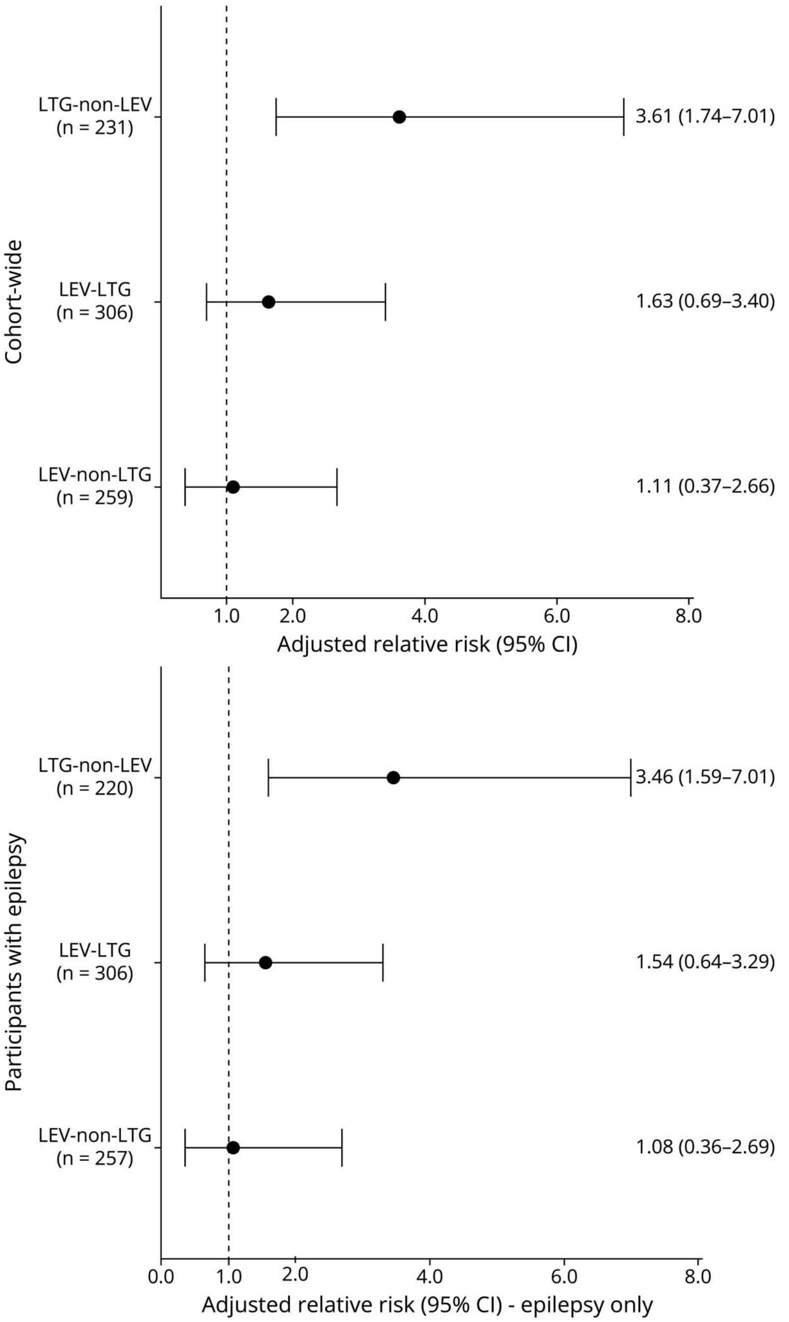
Data Availability

The study protocol and annual updates of tabulated data are available on the registry website (aedpregnancyregistry.org). Anonymized tabulated data not published within this article will be made available by request from any qualified investigator.

Results

From February 1st, 1997 through December 31st, 2024, a total of 796 ASM polytherapy-exposed pregnancies, 2,582

Figure 2 Adjusted Relative Risk of Major Congenital Malformations Identified Among Infants Who Had Been Exposed to Lamotrigine- or Levetiracetam-ASM Polytherapy Combination During the First Trimester Compared With the Active Comparator Group of Lamotrigine Monotherapy-Exposed Infants, in a Cohort-wide and an Epilepsy Participant-Only Analysis



Non-LTG and non-LEV ASMs include carbamazepine, lacosamide, topiramate, and zonisamide. % = percentage; CBZ = carbamazepine; LCM = lacosamide; LEV = levetiracetam; LTG = lamotrigine; RR = relative risk; TPM = topiramate; ZNS = zonisamide.

LTG monotherapy-exposed pregnancies (active comparator reference group), 1,389 LEV monotherapy-exposed pregnancies, and 517 TPM monotherapy-exposed pregnancies were enrolled and eligible for analysis (eFigure 1-). The average gestational age at enrollment was 16 weeks for the LTG monotherapy (reference) group, the LEV monotherapy group, the TPM monotherapy group, and the polytherapy group.

The most-reported ASM polytherapy combination was levetiracetam-lamotrigine (LEV-LTG) at 38.4% (n = 306) of polytherapy-exposed pregnancies. Of the analyzed polytherapy-exposed pregnancies, 71.0% (n = 565) contain LEV and 67.5% (n = 537) contain LTG (Table 1). Lamotrigine-carbamazepine (LTG-CBZ) was the most common duotherapy in the late 1990s and early 2000s but dropped off in frequency by 2005. From 2003 to 2006, use of LEV-LTG and LTG-topiramate

Table 4 Risk, Crude Relative Risk, and Adjusted Relative Risk of Major Congenital Malformations (MCM) Identified Among Infants Who Had Been Exposed to a Topiramate-Containing Polytherapy Combination During the First Trimester Compared With the Reference Group of Topiramate Monotherapy-Exposed Infants: North American AED Pregnancy Registry 1997–2024

	Cohort-wide (n = 659)				Participants with epilepsy only (n = 535)			
ASM regimen	Infant count, n	Major congenital malformations, n (%[a]) (95% CI)	Unadjusted RR (95% CI)	Adjusted RR[b] (95% CI)	Infant count, n	Major congenital malformations, n (%[a]) (95% CI)	Unadjusted RR (95% CI)	Adjusted RR[b] (95% CI)
TPM monotherapy	517	27 (5.2) (3.6–7.5)	Reference	Reference	403	26 (6.5) (4.4–9.3)	Reference	Reference
LEV-TPM	59	2 (3.4) (0.9–11.5)	0.64 (0.10–2.20)	0.50 (0.08–1.83)	58	2 (3.4) (1.0–11.7)	0.52 (0.08–1.80)	0.41 (0.06–1.52)
LTG-TPM	83	6 (7.2) (3.4–14.9)	1.41 (0.51–3.32)	1.70 (0.59–4.32)	74	5 (6.8) (2.9–14.9)	1.05 (0.35–2.62)	1.36 (0.42–3.74)

Abbreviations: % = percentage; LEV = levetiracetam; LTG = lamotrigine; n = sample number; RR = relative risk; TPM = topiramate.
[a] Percentage of infant with MCM count in each monotherapy or polytherapy group.
[b] Adjusting for maternal age, marital status, and calendar year of last menstrual period.

(LTG-TPM) duotherapies increased markedly. However, although steady use of LEV-LTG maintained until 2023, LTG-TPM duotherapy use dropped considerably by 2009 (-eFigure 2). Indication of ASM polytherapy combinations was largely epilepsy (98.4%), with minimal use for depression, migraines, mood disorder, and pain. In contrast, the reference LTG monotherapy group had an indication of epilepsy in a lower proportion of 81.4%. Self-reported dosage of each ASM in each ASM regimen is shown in Table 2. Within each polytherapy group, concomitant users were in the majority (72.9%–90.8%) compared with users that switched from 1 medication to another (eTable 3).

Demographic characteristics are presented in Table 1. The mean maternal age was 31.2 for the reference LTG monotherapy group, 30.9 for the LEV monotherapy group, 29.9 for the TPM monotherapy group, and 30.2 for the polytherapy group. A large majority of all groups identified as non-Hispanic White. The LEV-LTG group exhibited the most similar demographics profile to the LTG monotherapy group, except for proportion of epilepsy indication. Folic acid supplementation at conception was more common among the LTG monotherapy and LEV-LTG groups than groups of other polytherapy combinations. Among those with epilepsy, participants on polytherapy reported an earlier age of first seizure, hence earlier initiation of epilepsy, than those under monotherapy (LTG, LEV or TPM) regimens. Those on polytherapy were also more likely to have seizures during pregnancy (42.2%–62.4%) compared with those on monotherapy (LTG monotherapy: 25.0%; LEV monotherapy: 25.0%; TPM monotherapy: 32.8%) (Table 1).

Major Congenital Malformations

The risk of MCMs was 2.0% for the reference LTG monotherapy group and 1.9% for the LEV monotherapy group. In polytherapy combinations, risk of MCMs ranged from 1.3% to 7.2%. Compared with LTG monotherapy, the largest RR was observed for LTG-topiramate (LTG-TPM) duotherapy (crude RR: 3.79 [95% CI 1.42–8.45], adjusted RR: 4.54 [1.61–11.07]). The mean dosages of LTG were similar between LTG monotherapy and LTG-TPM duotherapy (eTable 2). The adjusted RRs of other polytherapy combinations range from 0.99 to 3.25 (Table 2, Figure 1), with CIs largely overlapping. Notably, other than the LTG-LEV duotherapy, the RRs for any other LTG-containing duotherapy appear similar to each other. The same is observed for any LEV-containing duotherapy. However, estimates were imprecise, with CIs compatible with a wide range of RRs. An analysis grouping LEV-containing duotherapies and LTG-containing duotherapies with reference LTG monotherapy showed adjusted RRs of 1.63 (95% CI 0.69–3.40) for LEV-LTG duotherapy, 1.11 (95% CI 0.37–2.66) for any LEV-containing duotherapy with an ASM aside from LTG (i.e., TPM, CBZ, lacosamide [LCM], zonisamide [ZNS]), and 3.61 (95% CI 1.74–7.01) for any LTG-containing duotherapy with an ASM aside from LEV (Table 3, Figure 2).

A comparison of LEV-TPM and LTG-TPM against TPM monotherapy (reference) for this study cohort found imprecise estimates of adjusted RRs of MCMs for both TPM-containing polytherapies: 0.50 (95% CI 0.08–1.83) and 1.70 (95% CI 0.59–4.32), respectively (Table 4).

Among participants with epilepsy, the direction and magnitude of RR of MCMs across all polytherapy groups were similar to the cohort-wide analysis (Tables 2 and 3, Figures 1 and 2). Our pure prospective analysis was limited because of small numbers and imprecise estimates (eTables 4 and 5), so all analyses included both the pure and traditional cohorts.

Small counts did not allow us to consider specific MCMs. However, among the infants exposed to any ASM polytherapy combination during their first trimester, 26 (3.3%) had MCMs and the most common were hypospadias (5 cases),

Table 5 Major Congenital Malformations Reported for Each Polytherapy Combination

Polytherapy combination	Participant record No./type of prospective exposure[a]	Birth status	Major congenital malformation[b]
LEV-CBZ (n = 76)	12,999/pure	Livebirth	Undescended testicle (right)
LEV-LCM (n = 65)	13,036/traditional	Livebirth	VSD, perimembranous
LEV-LTG (n = 306)	5,955/traditional	Livebirth	Malrotation of small intestine with surgery on first d of life
	6,297/pure	Livebirth	VSD, membranous (2.5 mm)
	8,934/traditional	Livebirth	VSD, perimembranous
	8,957/pure	Livebirth	ASD (5 mm)
	9,797/pure	Livebirth	Hypospadias, mid to proximal shaft location
	11,058/pure	Livebirth	Hypospadias
	18,613/traditional	Livebirth	Gastroschisis
	26,090/pure	Livebirth	VSD, large perimembranous; ASD, diameter > 5 mm
LEV-TPM (n = 59)	6,837/traditional	Livebirth	Hypospadias; VSD, perimembranous
	8,894/traditional	Livebirth	Cleft lip (unilateral, left) and palate
LEV-ZNS (n = 59)	25,397/traditional	Livebirth	Cleft lip (unilateral, left) and palate
LTG-CBZ (n = 85)	2,719/pure	Livebirth	One twin with esophageal atresia; renal agenesis (unilateral)
	3,334/pure	Elective termination	Myelomeningocele, lumbosacral
	6,013/traditional	Livebirth	ASD, small to medium in size
	26,434/pure	Livebirth	Inguinal hernia, unilateral
LTG-TPM (n = 83)	4,784/traditional	Livebirth	Severe chordee
	7,621/traditional	Livebirth	Chordee with surgery
	7,633/traditional	Livebirth	Inguinal hernia, left
	8,361/traditional	Livebirth	Cleft lip and palate, bilateral
	9,089/traditional	Livebirth	Hypospadias
	11,074/pure	Livebirth	Undescended testicle (left)
LTG-ZNS (n = 63)	12,338/traditional	Livebirth	Hypospadias
	18,928/pure	Livebirth	Cleft lip (unilateral, left) with intact palate
	25,899/traditional	Livebirth	Inguinal hernia, left

Abbreviations: ASD = atrial septal defect; CBZ = carbamazepine; LCM = lacosamide; LEV = levetiracetam; LTG = lamotrigine; TPM = topiramate; VSD = ventricular septal defect; ZNS = zonisamide.
[a] Pure and traditional types of prospective exposure are defined in the "Methods" section.
[b] Age of infant when malformation was detected was between birth and 5 days of age unless specified as an age between 5 days and 12 weeks.

ventricular septal defect (5 cases), and cleft lip/palate (4 cases). All cleft lip/palate cases occurred in combinations with either TPM or zonisamide (Table 5).

Discussion

We analyzed data from a cohort of women enrolled in the North American Anti-Epileptic Drug Pregnancy Registry (NAAPR) to assess the risk of MCMs in infants of participants taking ASM combinations (polytherapy) during the first trimester of pregnancy, compared with those taking LTG monotherapy. Our findings suggest an increased risk of MCM associated with LTG-TPM duotherapy. Relative risk estimates were too imprecise for us to conclude on the teratogenicity of other analyzed polytherapy combinations.

Of interest, our analyses suggest an increased risk of MCMs in LTG-containing duotherapies (without LEV) relative to LTG monotherapy, with results being most compatible with RRs between 1.7 and 7.0 based on the 95% CI. This was not observed for LEV-containing duotherapies (without LTG), although the 95% CI ranged from 0.37 to 2.66.

Specifically, we observed an increased risk of MCMs in those exposed to LTG-TPM polytherapy, concurrent with previous research demonstrating increased risk of MCMs from TPM in monotherapy[1,4,10,18-21] and polytherapy (compared with combinations without TPM or valproate)[11,22-26] during first trimester, specifically cleft lip and palate. Our results for LEV-TPM duotherapy were imprecise, although an increased risk has been suggested in previous literature.[27] Given higher rates of MCM observed in TPM monotherapy compared with LEV or LTG monotherapy,[10] we conducted additional analyses that yielded similar imprecise null effects of LEV-TPM and LTG-TPM compared with TPM monotherapy. Therefore, we cannot conclusively indicate that LEV-TPM and LTG-TPM are either safer or more teratogenic regimens than TPM monotherapy because of the wide CI and small sample size of polytherapy pregnancies.

Our sample sizes for the analyzed polytherapy combinations (excluding LEV-LTG) ranged from 59 to 85 pregnancies, suggesting that the study may have been underpowered to detect modest differences, especially in the context of an underlying rare prevalence of MCMs overall. Thus, the absence of statistical significance among polytherapy combinations of LEV-CBZ, LEV-LCM, LEV-TPM, LEV-ZNS, LTG-CBZ, and LTG-ZNS should be interpreted cautiously, as a type II error remains possible.

Few studies have investigated risk of MCM in newborns exposed to ASM polytherapy combinations in first trimester. The key strengths of this investigation include the comprehensive and longitudinal documentation of ASM use since 1996, forming a relatively large cohort of pregnancies of ASM users. Furthermore, MCM outcome definitions were assessed and confirmed by a teratologist review of medical records while blinded to exposure. This study also uses an active-comparator reference group of LTG monotherapy, shown to have risk of MFMs similar to the risk of pregnant women not exposed to any ASMs,[7,11] in the estimation of RR of MCMs, thereby comparing groups similar in treatment indications. To assess the role of ASM indication, our analysis further restricted to participants with epilepsy and produced similar magnitude and direction of RR of MCM.

This study does, however, have several limitations. First, MCMs in our study were assessed within 1 year after birth, potentially missing delayed MCM diagnoses. In addition, the observational nature of the registry allows for results susceptible to selection bias and confounding by unmeasured variables. Despite amassing close to 2 decades of registry data, most polytherapy combinations remain rare, and the small counts produce imprecise RR estimates, especially given the rarity of MCM occurrence in newborn. Infrequent use of certain ASM combinations precluded estimation of RRs for less common polytherapies, such as those including valproate. Furthermore, our study does not take into account temporality (i.e., concurrent or consecutive use) nor dosage of polytherapy combinations, restricting further investigation into dose-dependent relationships observed for ASMs such as valproate and TPM.[24,26,28-30] Our findings do, however, add to the literature of ASM polytherapy use and MCM risk and reinforce the significance of other studies to affirm our findings. Aforementioned small sample size limitations also underscore the importance of continuing monitoring the safety of ASM polytherapies through studies such as the NAAPR, in efforts to combat low sample sizes associated with polytherapy use.[31]

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Author Contributions
S.C.W. Ng: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. S. Hernandez-Diaz: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. M. Quinn: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. J.R. Ward: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. S. Conant: major role in the acquisition of data. A. Lyons: major role in the acquisition of data. F.A. High: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data.

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